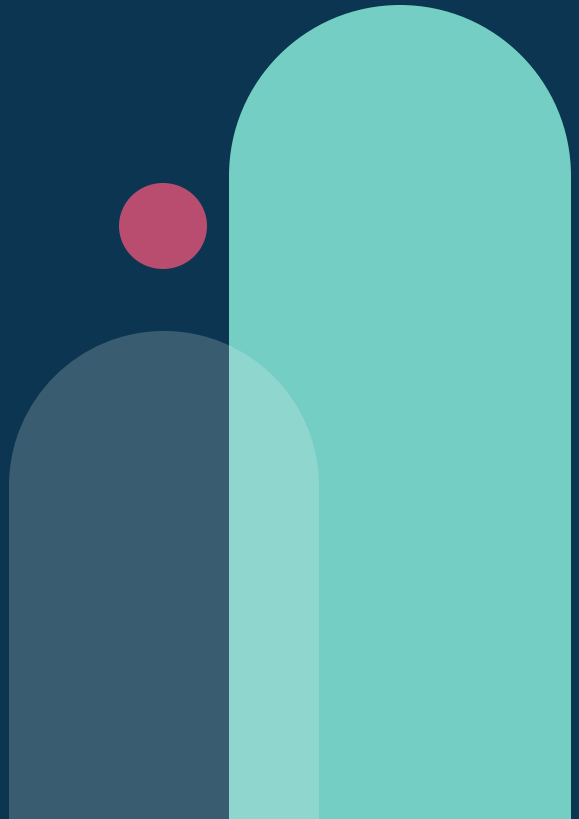
Prompt Engineering & Gradio

01

Prompt Engineering



LLM Settings

Temperature: Controls randomness; low for factual, high for creative tasks.

Top P: Nucleus sampling; low values select the most confident responses; high values introduce diversity.

Max Length: Sets a limit on token generation, controlling response length and cost.

Stop Sequences: Stops the model from generating further tokens, useful for structuring outputs.

Elements of a Prompt

Instruction: The specific task, e.g., “Classify the sentiment.”

Context: Additional information to improve responses.

Input Data: The subject of the task, e.g., “I think the food was okay.”

Output Indicator: Desired format of the result, e.g., “Sentiment: Positive.”

General Tips for Prompt Design

Start Simple: Iterate and refine over time for optimal results.

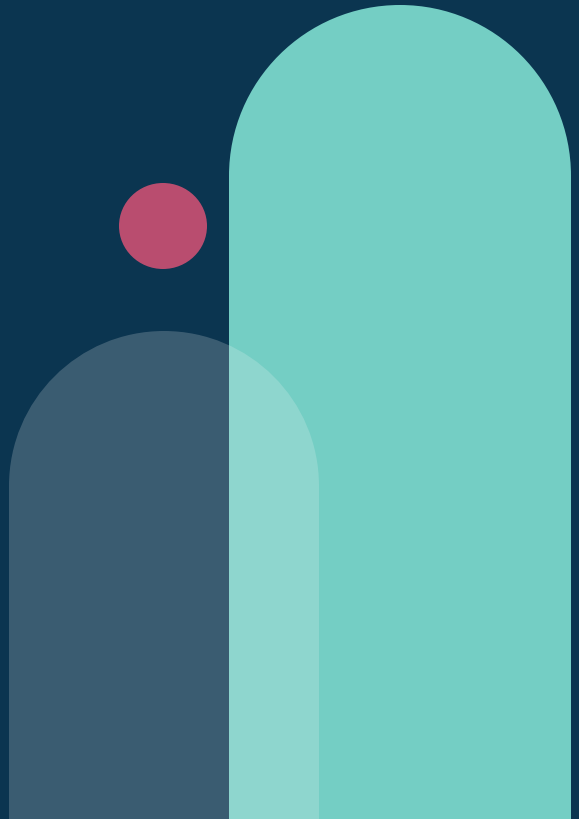
Be Specific: Detailed, clear instructions lead to better responses.

Avoid Imprecision: Direct, concise prompts work best.

Positive Framing: Focus on what to do, not what to avoid.

02

Prompting Techniques



Zero-Shot Prompting

Definition: The model performs tasks without prior examples or demonstrations.
Usage: Ideal for simple tasks or when the model is pre-trained on similar tasks.
Benefits: Efficient for straightforward tasks with minimal input.
Limitations: May struggle with complex tasks that require context or examples.

Prompt:

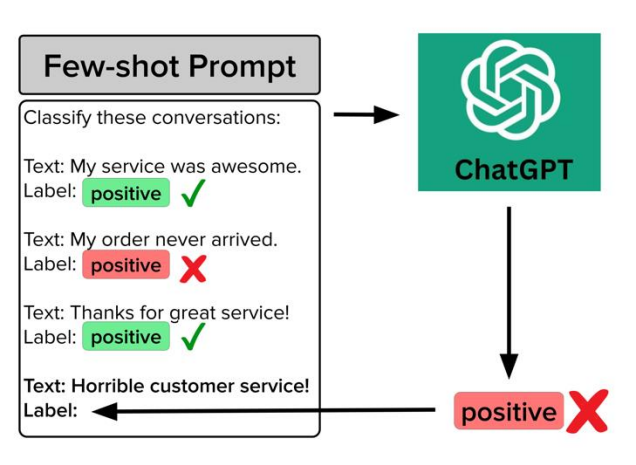
```
Classify the text into neutral, negative or positive.
Text: I think the vacation is okay.
Sentiment:
```

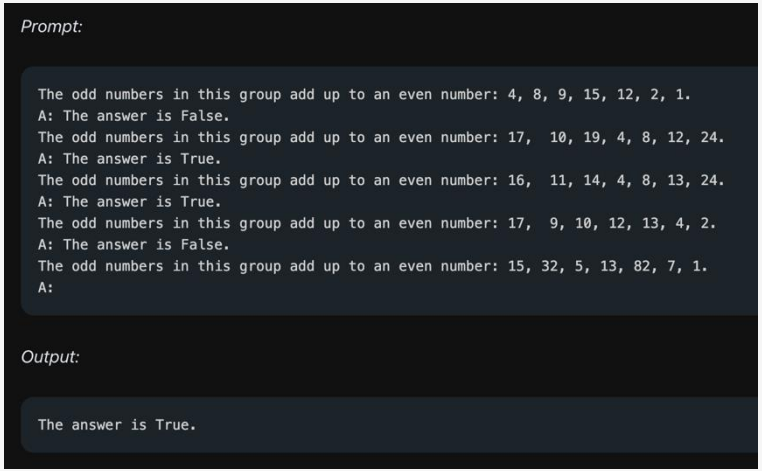
Output:

```
Neutral
```

Few-Shot Prompting

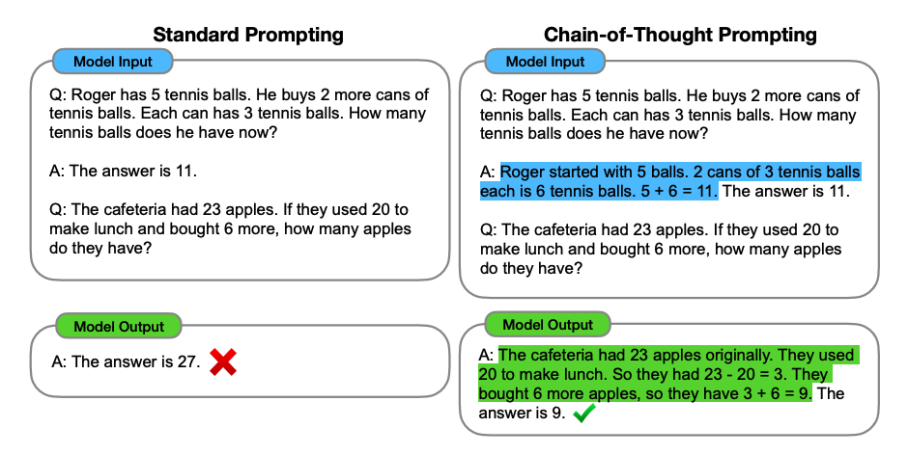
Definition: Providing a few examples (exemplars) to guide the model's output.
Usage: Useful for complex tasks where the model benefits from seeing relevant examples.
Benefits: Improves accuracy by adapting the model to specific tasks.
Limitations: Requires careful selection of examples and increases prompt length.





Chain of Thought (CoT) Prompting

Definition: Guides the model to break down reasoning tasks step-by-step.
Usage: Effective for tasks involving complex reasoning, such as multi-step problem solving.
Benefits: Encourages deeper reasoning and more accurate responses for complex tasks.
Limitations: Can lead to verbose outputs and is not suitable for every task.



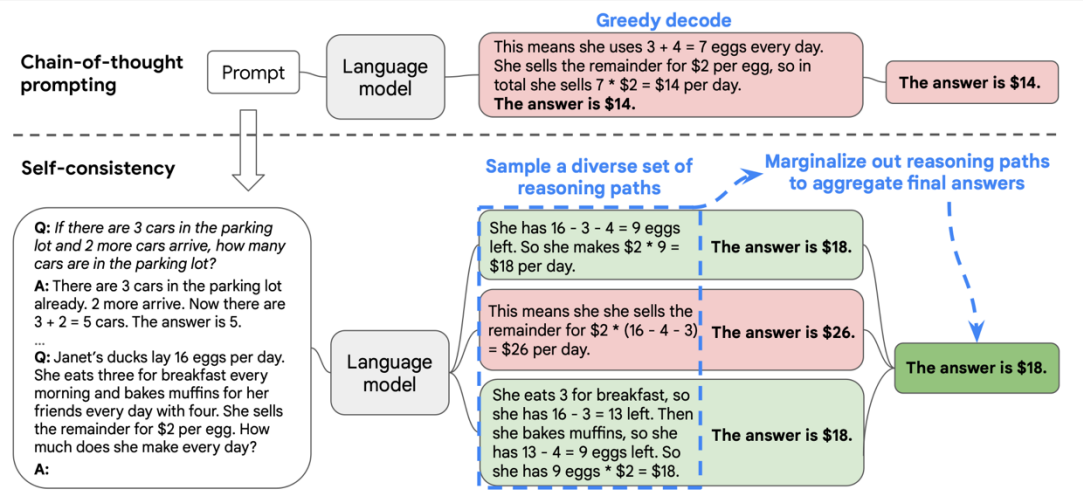
Self-Consistency

Definition: Generates multiple reasoning paths (CoT), then selects the most consistent result.

Usage: Helps improve accuracy for tasks that require detailed reasoning and minimizes errors.

Benefits: Increases robustness by selecting the most frequent or consistent response.

Limitations: Requires more computational resources to generate multiple outputs.



ReAct (Reasoning + Acting)

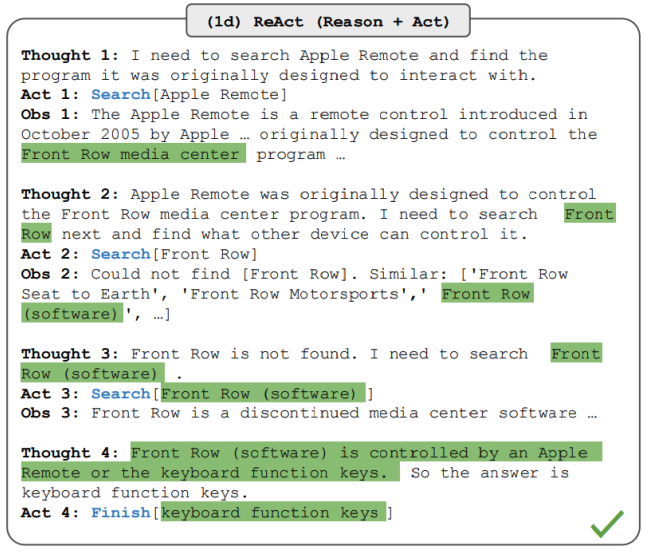
Definition: Combines reasoning with actions, allowing the model to interact with external tools or APIs.

Usage: Designed for tasks where the model needs to reason and act simultaneously.

Benefits: Ideal for scenarios requiring both logical reasoning and external interactions.

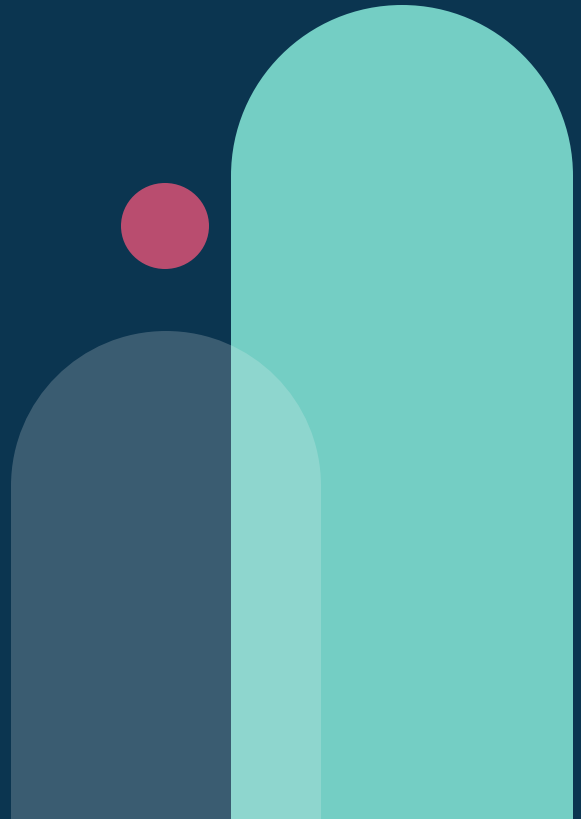
Limitations: Complex to implement and depends on tool availability.

Q: Aside from the Apple Remote, what other devices can control the program Apple Remote was originally designed to interact with?



03

Gradio



Gradio

- Gradio PlayGround
- Similarity Score
- Agent

Thanks!

Do you have any questions?

